Amer Rawashdeh

Department of Mathematics, Wayne State University, Detroit, MI 48202
rawashdeh@wayne.edu | amerry2015@gmail.com | Wayne State Profile | ORCID: 0009-0006-2937-2934

Research Interests

Nonlinear partial differential equations; liquid crystal flows; nonlinear analysis; variational and geometric methods; PDE models from continuum mechanics.

Education

Ph.D. in Mathematics, Wayne State University, Detroit, MI — Expected May 2027
Advisors: Fernando Charro and Tao Huang. GPA 3.91/4.00
Dissertation: One-dimensional models for the general Oseen–Frank energy: existence of weak and strong solutions

M.S. in Mathematics, Central Michigan University, Mount Pleasant, MI — May 2021
Advisor: Dmitry Zakharov. GPA 3.8/4.0
Plan B Project: Theory of valuations (after Neukirch, *Algebraic Number Theory*, Ch. II)

M.S. in Mathematics, Yarmouk University, Irbid, Jordan — September 2015
Advisor: Mowaffaq Hajja
Thesis: Number-theoretic theorems and conjectures in rings other than the integers, with emphasis on an analogue of Goldbach's conjecture in $\mathbb{Z}_n$

B.S. in Mathematics, Yarmouk University, Irbid, Jordan — August 2010

Publications

1. T. Huang, B. Rababah, and A. Rawashdeh, On existence of chemotaxis compressible Navier–Stokes equations modeling vascular network formation, *Journal of Differential Equations* **480** (2026), Article 114655. doi:10.1016/j.jde.2026.114655

Submitted Manuscripts

1. T. Huang, B. Rababah, and A. Rawashdeh, Compactness of the Ericksen–Leslie system modeling nematic liquid crystal flows in dimension two. Submitted September 2026.
2. T. Huang and A. Rawashdeh, Well-posedness of biaxial nematic flows with rough initial data. Submitted August 2026.
3. F. Charro and A. Rawashdeh, A characterization of the determinant in terms of elementary symmetric polynomials. Submitted June 2026.

Talks

1. On existence of chemotaxis compressible Navier–Stokes equations modeling vascular network formation. SIAM Great Lakes Section Annual Meeting, Purdue University, September 2026.
2. Tensor products in algebra and in PDE, and the Einstein notation. Informal PDE Seminar, Wayne State University, May 2025.

3. Hausdorff measure and dimension, and the area and coarea formulas. Special Seminar, Wayne State University, May 2024.

Teaching Experience

Wayne State University, Detroit, MI 2022–present
Graduate Teaching Assistant; Instructor of Record for each course listed

MAT 1010, Beginning Algebra	Fall 2026
MAT 2020, Calculus II	Spring/Summer 2026; Winter 2026; Spring/Summer 2023
MAT 2150, Differential Equations and Matrix Algebra	Fall 2025
MAT 2250, Elementary Linear Algebra	Spring/Summer 2025; Winter 2025
MAT 2010, Calculus I (coordinated flipped-classroom format)	Fall 2024
STA 2210, Probability and Statistics	Spring/Summer 2024
MAT 2030, Calculus III	Winter 2024
MAT 0993, Beginning Algebra (coordinated lab-based format)	Fall 2023
MAT 1800, Elementary Functions	Winter 2023

- Tutor, Mathematics Resource Center and Mathematics Computer Lab.

Central Michigan University, Mount Pleasant, MI 2018–2021
Teaching Assistant and Research Assistant
- Instructor of record, MTH 105, College Algebra (Fall 2020, Spring 2021).
- Tutor, Mathematics Assistance Center; Outstanding Tutoring Award (2021–2022).

Ministry of Education, Irbid, Jordan 2010–2018
Mathematics Teacher
- Hitteen Secondary School (2015–2018): Grade 12 mathematics including calculus; Grade 11 mathematics including precalculus.
- Kufr-Asad Secondary School (2010–2015): Grades 8–11 algebra, analytic and synthetic geometry, functions, introductory statistics, and probability.

Curriculum and Instructional Development

- **Calculus I flipped-classroom course**, Fall 2024. Taught a section in the first semester of the department's coordinated flipped Calculus I; attended weekly course-team meetings and submitted written recommendations on course materials to the department.
- **Analysis Preliminary Exam preparation**, Fall 2023. Co-developed solutions and study materials for graduate students preparing for the Analysis Preliminary Examination.
- **Tawjihi Mathematics Study Guide**, 2012. Wrote a study guide for the mathematics component of Jordan's national secondary-school examination.

Honors and Awards

William Martin Borgman Endowed Scholarship for Mathematics, Wayne State University	2026–2027
M. F. Janowitz Scholarship, Wayne State University	2023–2024
Outstanding Tutoring Award, Graduate Student, Central Michigan University	2021–2022

Service

Co-organizer, AMS Special Session on Recent Trends in Nonlinear PDEs and Applications, April 2027

AMS Spring Eastern Sectional Meeting
Teaching Subcommittee, Mathematics Strategic Planning Committee, Wayne State University Winter 2026
President, SIAM Student Chapter, Wayne State University Fall 2025–Winter 2026
Organizer, Informal PDE Seminar, Wayne State University Spring/Summer 2025
Student Representative, Department Chair Review Committee, Wayne State University 2024–2025
Participant, Inclusive Excellence Listening Session, Wayne State University Winter 2026
Conference support, SIAM Great Lakes Section Meeting, Wayne State University September 2022

Professional Memberships

American Mathematical Society (AMS) 2022–present
Society for Industrial and Applied Mathematics (SIAM) 2024–present

Languages

Arabic (native); English (fluent)